must lift a little weight off the left pan. On the other hand, we could lift a one-unit weight by lowering the three-unit weight, if we cheat a little by lifting a little weight off the other pan. Of course, we realize that with any *actual* lifting machine, we must add a little extra to get it to run. This we disregard, *temporarily.* Ideal machines, although they do not exist, do not require anything extra. A machine that we actually use can be, in a sense, *almost* reversible: that is, if it will lift the weight of three by lowering a weight of one, then it will also lift nearly the weight of one the same amount by lowering the weight of three.

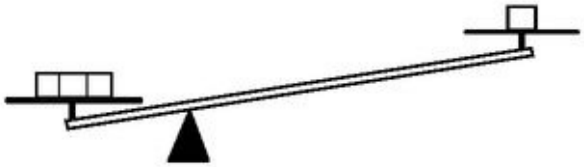


Figure 4-1 Simple weight-lifting machine.

We imagine that there are two classes of machines, those that are *not* reversible, which includes all real machines, and those that *are* reversible, which of course are actually not attainable no matter how careful we may be in our design of bearings, levers, etc. We suppose, however, that there is such a thing—a reversible machine—which lowers one unit of weight (a pound or any other unit) by one unit of distance, and at the same time lifts a three-unit weight. Call this reversible machine Machine *A.* Suppose this particular reversible machine lifts the three-unit weight a distance *X.* Then suppose we have another machine, Machine *B,* which is not necessarily reversible, which also lowers a unit weight a unit distance, but which lifts three units a distance *Y.* We can now prove that *Y* is not higher than *X;* that is, it is impossible to build a machine that will lift a weight *any higher* than it will be lifted by a reversible machine. Let us see why. Let us suppose that *Y* was higher than *X.* We take a one-unit weight and lower it one unit height with Machine *B,* and that lifts the three-unit weight up a distance *Y.* Then we could lower the weight from *Y* to *X, obtaining free power,* and use the reversible Machine *A,* running backwards, to lower the three-unit weight a distance *X* and lift the one-unit weight by one unit height. This will put the one-unit weight back where it was before, and leave both machines ready to be used again! We would therefore have perpetual motion if *Y* were higher than *X,* which we assumed was impossible. With those assumptions, we thus deduce that *Y is not higher than X,* so that of all machines that can be designed, the reversible machine is the best.